

BOOK IV — CERTIFICATION PROOF PACK

Document ID: PH6-BOOKIV-CONSOLIDATED-1.0

Role: CERTIFICATION

Status: ACTIVE

Question Answered: How truth is proven.

B4 § 1 — Certification Law

Certification verifies doctrine; certification does not modify doctrine.

B4 § 2 — Schema Contracts

- **EvidencePacket Schema:** (Frozen dataclass, PASS/DROP only, canon_locked=True)
- **Receipt Schema:** (ph6_receipt_v1, monotonic event_seq, event_hash, authority_hash)
- **Replay Manifest Schema:** (ph6_replay_v1, expected_verdict, expected_metrics)
- **Audit Event Schema:** (ph6_audit_v1, all 8 mandatory fields)

B4 § 3 — Hash and Serialization Doctrine

- **BLAKE2b-256 Contract:** Primary evidentiary hash.
- **Canonical JSON Rules:** sort_keys=True, ensure_ascii=False, allow_nan=False.

B4 § 4 — C1 Golden Vectors

Canonical determinism conformance test set. verdict, metrics, and result_set_hash must match exactly.

B4 § 5 — HRG9 Evidence Gate

Status: OPEN / EVIDENCE BLOCKED

Required for closure:

- P1/P2/P3 vector parity PASS.
- result_set_hash parity across all three.
- authority_hash verification PASS.

- hrg9_manifest.json present.
- hrg9_replay_parity_receipt.json emitted.

B4 § 6 — Certification Matrix

1. EvidencePacket schema conformance
2. Replay determinism proof
3. BLAKE2b-256 hash stability
4. Canonical JSON conformance
5. Atomic write power-loss safety
6. Lane boundary isolation
7. HRG9 evidence gate closure

B4 § 7 — Audit Implementation Compliance

`append_audit()` must emit: `schema` , `event_seq` , `event_type` , `object_id` , `event_hash` , `prev_event_hash` , `authority_hash` , `ts` .

B4 § 8 — Certification Workflow

Pre-certification checklist and status labels (CANON, CERTIFICATION, EXPERIMENTAL, SUPERSEDED, DRAFT).